

Non-calculator Part TURN IN THIS SHEET BEFORE USING YOUR CALC ON PAGE 3

1. Consider the set of parametric equations $x = 1 + \frac{2}{t-3}$, $y = t - 2$ for $-1 \leq t \leq 4$.

a) Complete the table of values, sketch the graph, and determine the domain. [6 pts]

t	-1	1	2	2.5	3.5	4
x	$\frac{1}{2}$	0	-1	-3	4	3
y	-3	-1	0	$\frac{1}{2}$	$\frac{3}{2}$	2

-0 Domain: $(-\infty, 1) \cup (1, \infty)$

b) Eliminate the parameter and express y as a function of x . [3pts]

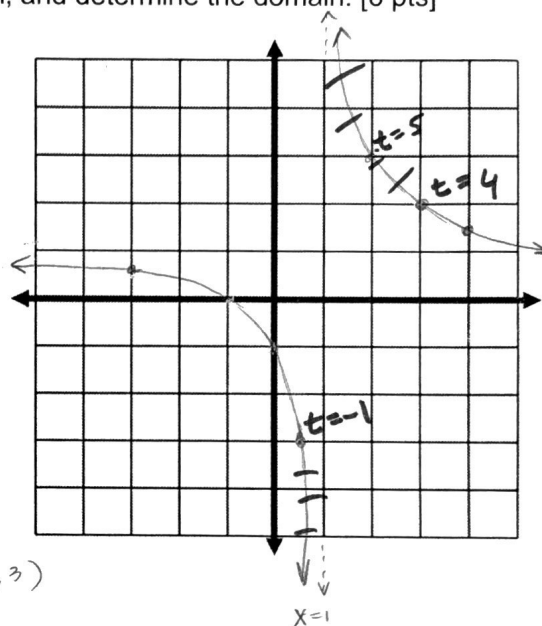
$$t = y + 2$$

$$x = 1 + \frac{2}{y+2-3} = 1 + \frac{2}{y-1}$$

$$x-1 = \frac{2}{y-1} \quad y-1 = \frac{2}{x-1}$$

$$y = \frac{2}{x-1} + 1$$

$$y = \frac{2}{x-1} + 1 \quad \checkmark$$



(2, 3)

$$0 = \cos 3t$$

$$3t = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$t = \frac{\pi}{6}, \frac{3\pi}{6} = \frac{\pi}{2}$$

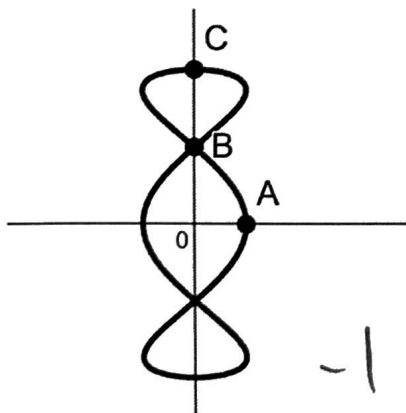
$$3 \sin\left(\frac{\pi}{6}\right) = \frac{1}{2} \times 3$$

$$3 \sin\left(\frac{\pi}{2}\right) = 3$$

2. The set of parametric equations $x = \cos(3t)$, $y = 3\sin t$ for $0 \leq t < 2\pi$ is graphed below.

a) Complete the following table of values about points A, B, and C labeled on the graph. [3 pts]

Point	A	B	C
x	1	0	0
y	0	$\frac{3}{2}$	3



b) What values of t draw the loop that starts and ends at point B and passes through point C? [2 pts]

$$x = \cos 3t$$

$$y = 3 \sin t$$

$$0 = 3 \sin t$$

$$\sin t = 0$$

$$t = 0, 2\pi$$

$$\cos(0) \cos(6\pi)$$

$$0 = \cos 3t$$

$$3t = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$t = \frac{\pi}{6}, \frac{3\pi}{6}$$

$$\Rightarrow y = 3 \sin\left(\frac{\pi}{6}\right) \rightarrow y = 3\left(\frac{1}{2}\right) = \frac{3}{2}$$

$$y = 3 \sin\left(\frac{\pi}{2}\right) \rightarrow y = 3$$



$$\frac{\pi}{2} - \frac{\pi}{6}$$

$$\frac{\pi}{6} < t < \frac{\pi}{2}$$

3. Each set of parametric equations (i) – (iii) converts to the same function: $y = x^4 - x^2$. Write letters A – D to match the parametric equations with their graphs. Letters may repeat. Not all letters will be used. No work is required to be shown. [1pt each]

(i) $x = \sin(t), y = -\sin^2(t)\cos^2(t)$

A ✓ $D: [-1, 1]$
 $R:$

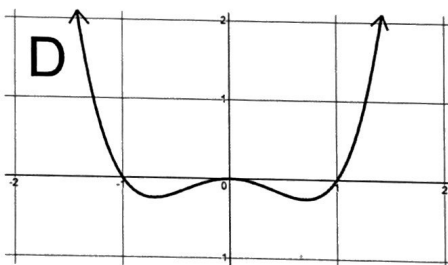
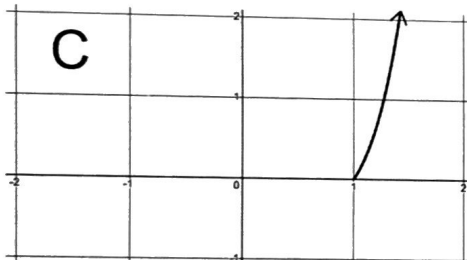
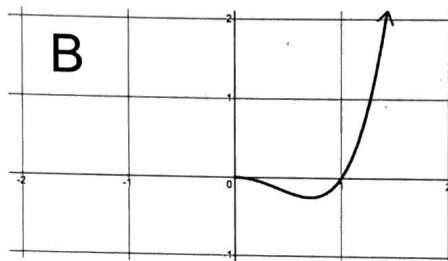
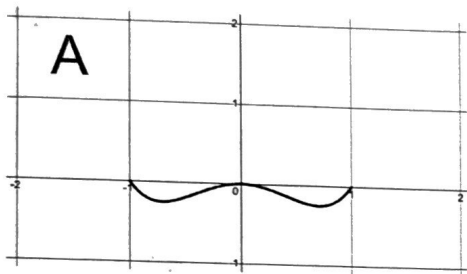
(ii) $x = \frac{1}{\sqrt{1-t}}, y = \frac{t}{(1-t)^2}$

B ✓ $D: (0, \infty)$
 $R:$

(iii) $x = t, y = t^2(t^2 - 1)$

D ✓ $D: \mathbb{R}$
 $y:$

$y = x^2(x^2 - 1)$
 $y = x^4 - x^2$



4. Consider the vectors $\vec{u} = \langle -1, 3 \rangle$ and $\vec{p} = \langle 2, -5 \rangle$. [9 pts]

a) $|\vec{u}|$

$\sqrt{10}$ ✓

d) $\hat{u}$ (a unit vector in the direction of $\vec{u}$)

$\langle \frac{-1}{\sqrt{10}}, \frac{3}{\sqrt{10}} \rangle$ ✓

b) Write $\vec{u}$ as a linear combination of $\hat{i}$ and $\hat{j}$

$-1\hat{i} + 3\hat{j}$ ✓

e) Calculate $\vec{u} \cdot \vec{p}$

$(-1)(2) + (3)(-5)$
 $-2 - 15$

-17 ✓

c) $\vec{u} - 2\vec{p}$

$\langle -1, 3 \rangle - \langle 4, -10 \rangle$

$\langle -5, 13 \rangle$ ✓

f) Find a unit vector orthogonal to $\vec{p}$

$2(x) + (-5)y = 0$

$\langle \frac{5}{\sqrt{29}}, \frac{2}{\sqrt{29}} \rangle$ ✓

- g) For what values of k is the angle between vectors $\langle k, 3 \rangle$ and $\vec{p}$ acute?

$\langle k, 3 \rangle \cdot \langle 2, -5 \rangle$

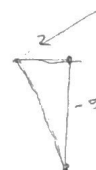
$2k - 5(3) > 0$

dot product > 0

$2k - 15 > 0$

$2k > 15$

$k > \frac{15}{2}$ ✓



$\frac{25}{49}$

$m: \sqrt{29}$
 $\langle 5, 2 \rangle$

$2x - 5y = 0$

Hlasek / Deggeller!

Vectorious vectorizer: elle orang

AH 25/26

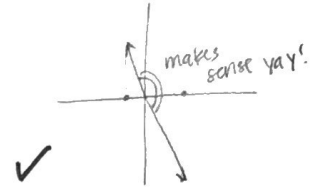
Unit 6 Quiz: Vectors and Parametrics

Period: 5

Calculator Part: Show work. Write on paper what you type into your calculator.

5. Find the angle between vectors $\vec{u} = \langle -1, 3 \rangle$ and $\vec{v} = \langle 2, -5 \rangle$. Give your answer rounded to the nearest hundredth of a degree. [2pts]

$$\cos^{-1} \left(\frac{(-1)(2) + (3)(-5)}{\sqrt{10} \cdot \sqrt{29}} \right) = \cos^{-1} \left(\frac{-17}{\sqrt{290}} \right) = \boxed{176.63^\circ}$$



6. Mrs. Hlasek is practicing long distance throwing in preparation for Game of Throws. She throws a juggling ball with velocity 30 feet per second at an angle 35° with initial height 3 ft.

$v = 30$
 $\theta = 35$
 $h_0 = 3$

- a) How long does it take for the juggling ball to reach its highest point? Give your answer rounded to the nearest hundredth of a second. [2pts]

$\frac{-b}{2a}$ ✓

$$x = 30t \cos 35 \quad \checkmark \quad 30t = \frac{x}{\cos 35} \quad y = \frac{-16}{(30 \cos 35)^2} x^2 + \tan 35 x + 3$$

$$y = 30t \sin 35 + 3 - 16t^2 \quad \checkmark \quad t = \frac{x}{30 \cos 35} \quad \frac{-b}{2a} = \frac{-\tan 35}{2 \left(\frac{-16}{(30 \cos 35)^2} \right)} = 0.440481$$

$$y = \frac{x}{\cos 35} \cdot \sin 35 + 3 - 16t^2 \quad \frac{-b}{2a} = \frac{-\tan 35}{2 \left(\frac{-16}{(30 \cos 35)^2} \right)} = 0.440481$$

$$y = \tan 35 x + 3 - 16 \left(\frac{x}{30 \cos 35} \right)^2$$

$15.0903 = 30t \cos 35$
 $t = 0.614064$

$t = 0.61$

- b) How far away from Mrs. Hlasek will the juggling ball hit the ground? Give your answer rounded to the nearest hundredth of a foot. [3pts]

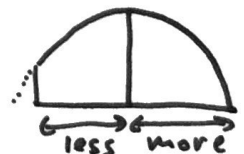
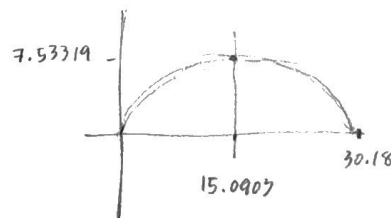
when $y=0$, what is x ? (and $t \neq 0$)

$$0 = 30t \sin 35 + 3 - 16t^2$$

$$16t^2 - t(30 \sin 35) - 3 = 0$$

$$t = \frac{30 \sin 35 \pm \sqrt{(30 \sin 35)^2 - 4(16)(-3)}}{2(16)} = 1.22813$$

$$x = 30(t) \cos 35 = \boxed{30.18} \text{ ft}$$



what is t when

$x = \frac{-b}{2a}$?

7.53319